MIL-C-50871C(AR)
22 JULY 1981
SUPERSEDING
MIL-C-50871B(PA)
27 April 1973

MILITARY SPECIFICATION
CARTRIDGE, 40MM, HEDP, M433; PROJECTILE METAL PARTS FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers, the metal parts for one type of cartridge designated as Cartridge, 40MM, HEDP, M433.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-C-5541	-	Chemical Film and Chemical Film Materials for Aluminum and Aluminum Alloys

STANDARD

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attrubutes
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1188	-	Commerical Packaging of Supplies and Equipment

FSC: 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKING DRAWINGS

8886375	-	Cup and Skirt Assembly
8886340	-	Ring, Retaining

INSPECTION EQUIPMENT DRAWINGS

8870392	-	Ultrasonic Test Equipment

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 117	-	Standard Method of Salt Spray (For Testing)

(Application for copies should be addressed to: American Society for testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawing (Dwg. 8886375 and 8886340), all associated drawings, and with all requirements specified in applicable specifications.

3.3 First article testing. This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

3.4 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharpy defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and quality assurance provisions Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are made a part of this detail specification.

4.2 Classification of inspections. The following type of inspections shall be conducted on this item:

a. First article inspection.
b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Cup and Skirt Assembly	8886375	50
Skirt,	8886344	50
Boattail Cup	9276546	20
Retaining Rings	8886340	50 (25 prior to heat treatment)
Embossed Stock	9276546	2 samples width of coil by 20 inches prior to forming the cup

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Projectile Components				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup and skirt assembly (Dwg. 8886375)				
	Examination for defects	50			4.4.2.4
	Air pressure test	50			4.5.2
	Push out test	50			4.5.3
	Skirt (Dwg. 8886344)				
	Examination for defects	50			4.4.2.1
	Salt spray test	4			4.5.5
	Cup, Boattail (Dwg. 9276546)				
	Examination for defects	50			4.4.2.2
	Retaining Ring (Dwg. 8886340)				
	Examination for defects	25			4.4.2.3
	Push out test (after heat treatment)	25			4.5.4

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50871C(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8886344-1
4.4.2.1	Skirt				NEXT HIGHER ASSEMBLY 8886375
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Distance between fuze seating surface and cup seating surface		0.40%	3.2	Gage
102.	Concentricity of bourrelet		0.40%	3.2	Gage
103.	Diameter rotating band		0.40%	3.2	Gage
104.	Salt spray test	*	*		4.5.4
105.	Diameter of fuze cavity		0.40%	3.2	Gage
MINOR					
201.	Total length		0.65%	3.2	Gage
202.	Diameter rear of rotating bands		0.65%	3.2	Gage
203.	Diameter of cup cavity		0.65%	3.2	Gage
204.	Smallest outside diameter		0.65%	3.2	Gage
205.	Diameter between rotating bands		0.65%	3.2	Gage
206.	Evidence of poor workmanship		0.65%	3.4	Visual
207.	Incorrect color		0.65%	3.2	Visual

NOTES:

* Sampling and rejection shall be in accordance with MIL-C-5541.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50871C(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Cup, Boattail		9276547
			NEXT HIGHER ASSEMBLY 8886375

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Presence of cracks, voids, pipes or inclusion, etc		100%	3.2	8870372 (4.5.1)
102.	Outside dia, max.	See 4.4.3.1	0.40%	3.2	Gage
103.	Total length, max.		0.40%	3.2	Gage
104.	Location of grooves		0.40%	3.2	Gage
105.	Depth of grooves		0.40%	3.2	Gage
106.	Width of grooves		0.40%	3.2	Gage
MINOR					
201.	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Retaining Ring, Prior to Heat Treatment	1 OF 1	8886340
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined	See 4.4.3.1			
MAJOR					
101.	Inside diameter		0.40%	3.2	Gage
102.	Outside diameter		0.40%	3.2	Gage
103.	Concentricity of inside diameter with outside diameter		0.40%	3.2	Gage
MINOR					
201.	Total height		0.65%	3.2	Gage
202.	Protective finish with bare spot		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50871C(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Cup and Skirt Assembly		8886375
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR					
101.	Bourrelet diameter		0.40%	3.2	Gage
102.	Rotating bands diameter, max.		0.40%	3.2	Gage
103.	Run out of boattail surface with fuze cavity including perpendicularity with fuze cavity depth		0.40%	3.2	Gage
			0.40%	3.2	Visual
104.	Adhesive not 360°		0.40%	3.2	Gage
105.	Length skirt to rear rotating band, max.		100%	3.2	4.5.2
106.	Air pressure test	32/ shift	*	3.2	4.5.3
107.	Push out test				
MINOR					
201.	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

* If any retaining ring fails the drawing requirement, the shifts preoduction represented by the sample shall be rejected.

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4.4.3 Testing.

4.4.3.1 Dimensional control of stamped parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.4.3.2 Dimensional control of embossed stock. At the start of production and for every four (4) hours production thereafter, prior to forming a cup, samples (width of coil by 20 inches), shall be selected from a coil of embossed stock representative of that four hours production. One sample strip shall be taken from the beginning and the end of the coil. The sample strips shall be identified as to their being the beginning or end of the coil and also the direction of milling. If the average value for fifteen (15) measuements for the depth of depression and embossed material thickness of the sample fails to comply with the requirement specified on the applicable drawing corrective action shall be taken prior to further embossing.

An additional sample will be selected at the introduction of each new embossing roll to the production line. If the average value for fifteen (15) measurements for depth of depression, embossed material thickness and center distance of embossment of the sample fails to comply with the requirement specified on the applicable drawing, the embossing roll will not be used for production and corrective action taken.

4.4.3.3 Push out test of retaining rings. Thirty-two (32) retaining rings shall be selected from each shift's production and subjected to this test. If one (1) or more retaining rings fail to comply with the requirement specified on the applicable drawing, the shift's production represented by the sample shall be rejected.

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4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method Column and the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Ultrasonic inspection. Cup, boattail shall be inspected for the presence of cracks, voids, pipes, or inclusions, etc., along the cylindrical section of the cup throughout the entire wall thickness including base section, gaged externally to a depth as specified on applicable drawings. This test shall be performed and calibrated as shown on the equipment drawing. If calibration shows equipment not to be performing correctly, the quantity represented prior to calibration shall be reprocessed.

4.5.2 Air pressure test of cup and skirt assembly. The cup and skirt assembly shall be placed in an approved fixture and the required air pressure specified on the applicable drawing shall be applied. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.5.3 Push out test of cup and skirt assembly. The cup and skirt assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.4 Push out test of retaining ring (after heat treatment). The retaining ring shall be replaced in an approved fixture and the push out force specified on the applicable drawing shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.5 Salt spray of skirt. This test shall be conducted in accordance with ASTM B 117 except as noted on applicable drawing. This test is a destructive test. Parts so tested shall not be returned to the lot.

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4.5.6 Dimensional inspection of embossed stock. Fifteen (15) measurements shall be conducted on the sample strip for each characteristic. These shall be three (3) points of measurements made across the width of the sample strip at five points along the length of approximately every four inches. One measurement for each characteristic shall be maintained at each measuring point across the width of the material. An average value of each sample shall be completed for each characteristic from data obtained.

5. PACKAGING

5.1 Preservation and packaging. Unless otherwise specified, packaging shall be Level B. Maximum and intermediate protection (Level A and B) shall be in accordance with MIL-STD-1169. Minimum protection shall be in accordance with MIL-STD-1188.

5.2 Unitization. Unitization for Levels A and B shall be in accordance with MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. This specification covers the metal parts, assembling and packing for Cartridge, 40MM, HEDP, M433.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment design for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAA-I, Dover, NJ 07801. Request letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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MIL-C-50871C Cartridge, 40MM, HEDP, M433, Projectile Metal PArts For

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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